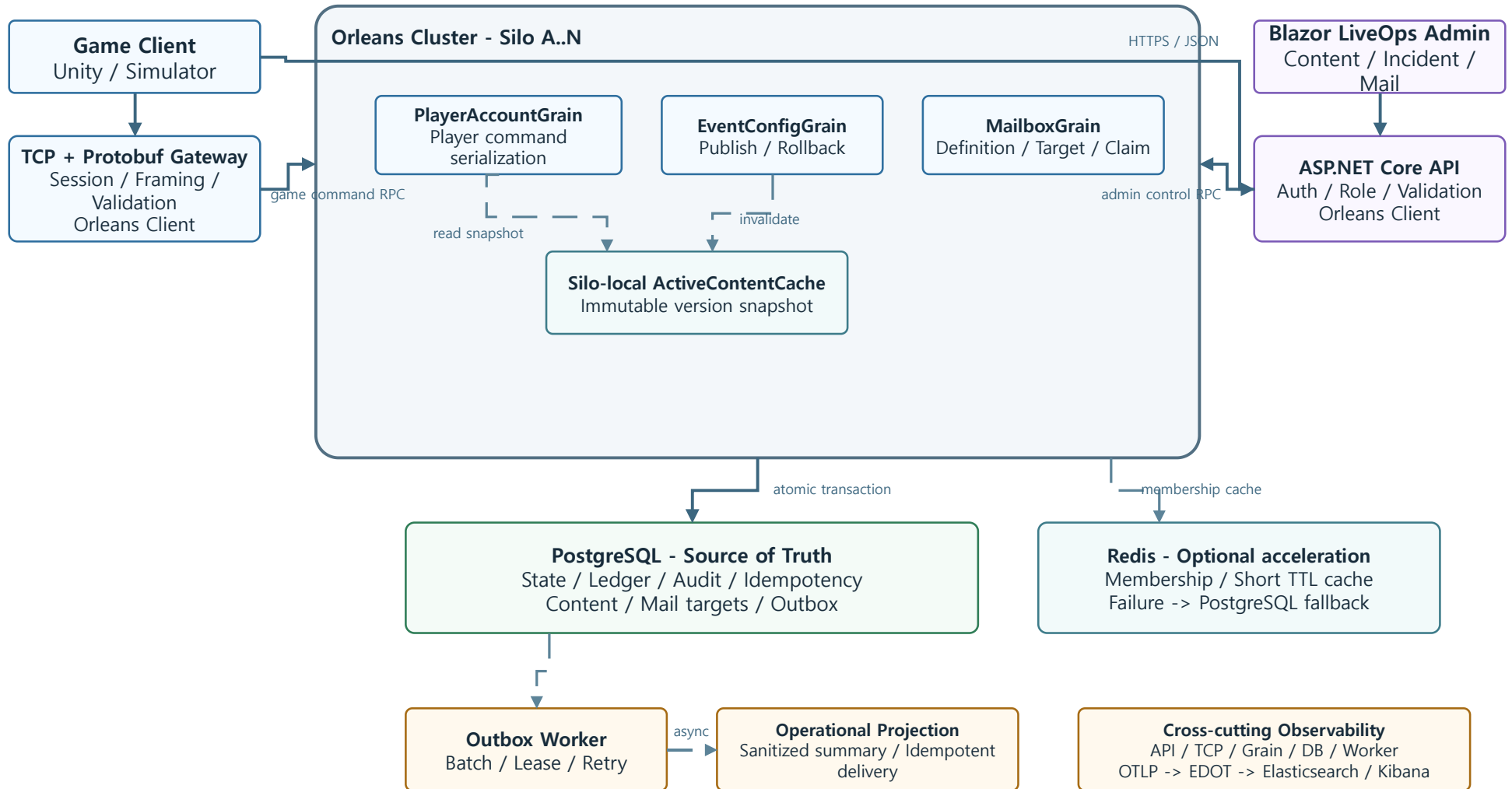


## 8.1 전체 구성

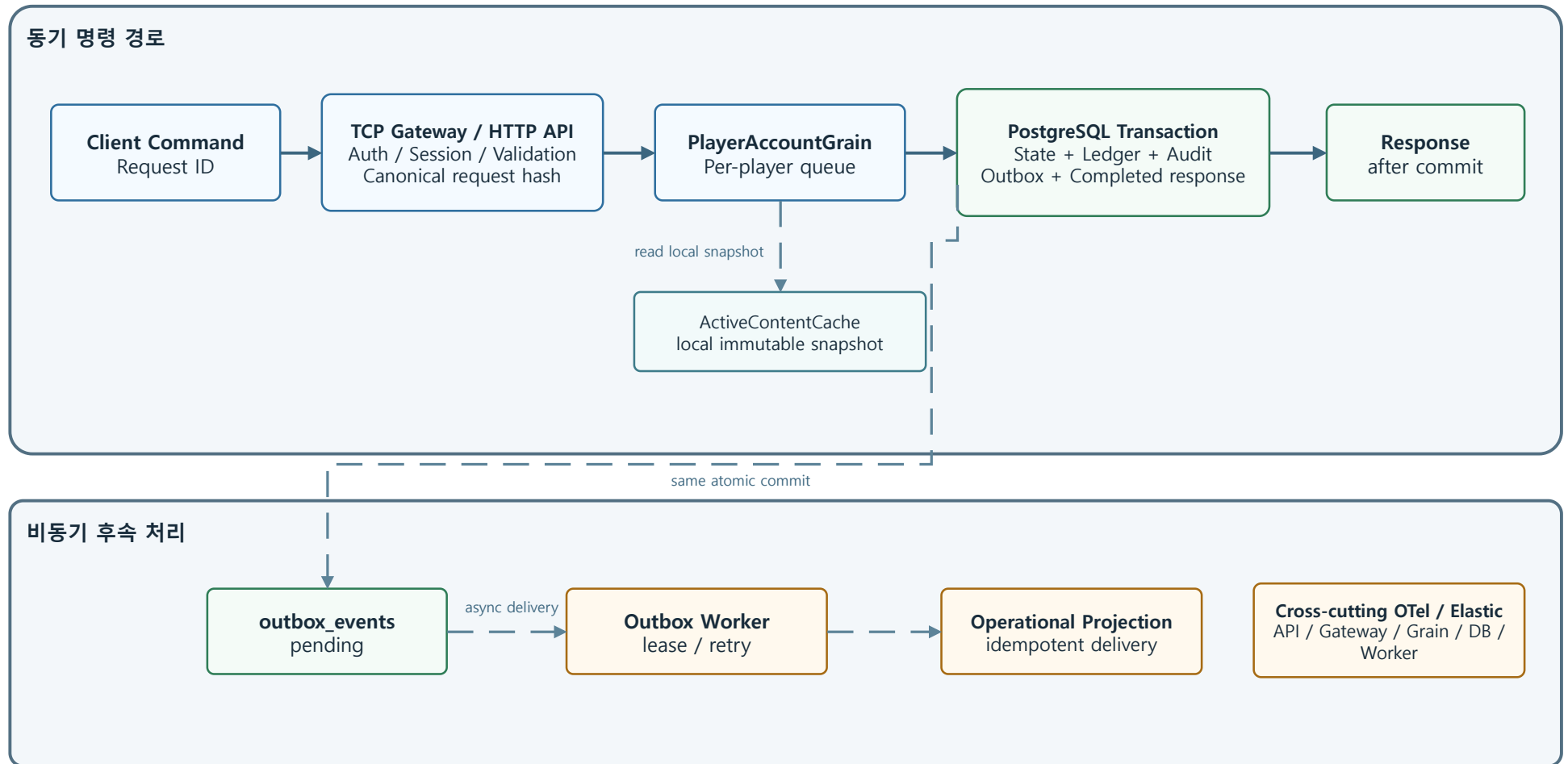
게임 명령 경로와 운영 제어 경로는 분리하고, PostgreSQL commit 이후 비동기 처리를 분기한다.



Solid: command / transaction | Dashed: cache / async | Observability is cross-cutting, not an Outbox destination

## 8.2 동기 요청과 비동기 분기

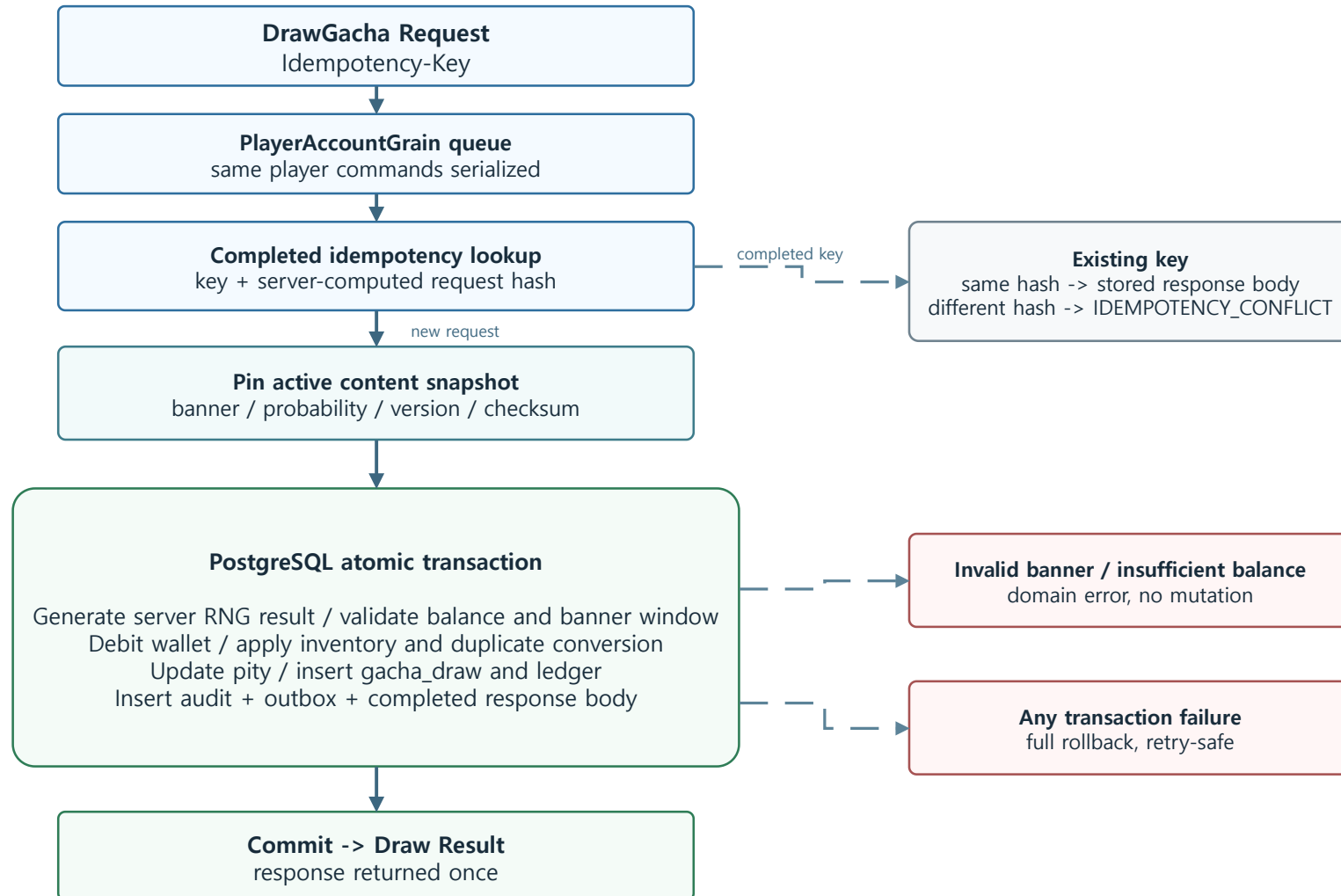
Worker와 Observability를 동기 요청의 직렬 단계로 오해하지 않도록 실행 경계를 분리한다.



The synchronous response path ends at PostgreSQL commit; asynchronous delivery failures do not roll back player state.

## 8.3 가차 원자 트랜잭션

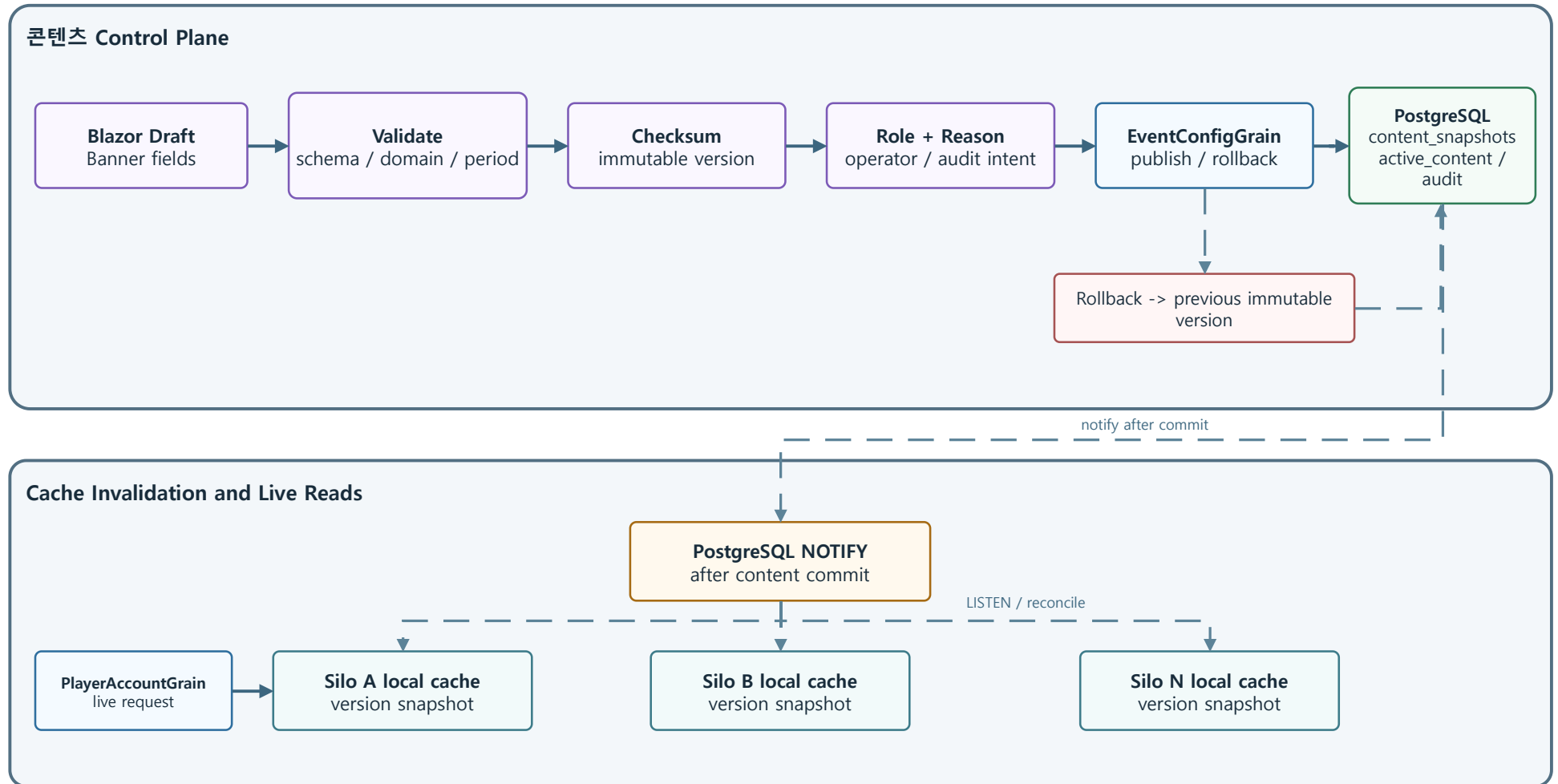
가차 결과, 재화, 인벤토리, 천장, 이력, Outbox, 먹등성 응답을 하나의 commit으로 묶는다.



Completed replay returns before reading current content; request idempotency stores no PENDING lock.

## 8.4 콘텐츠 배포와 Silo 캐시

EventConfigGrain은 control plane에만 두고, live request는 각 Silo의 local snapshot을 읽는다.

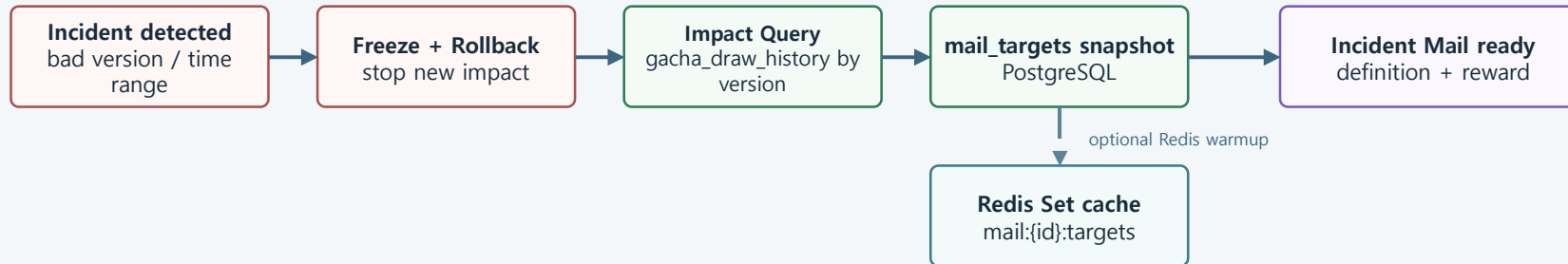


Hot path: PlayerAccountGrain -> local cache. It does not call EventConfigGrain for every request.

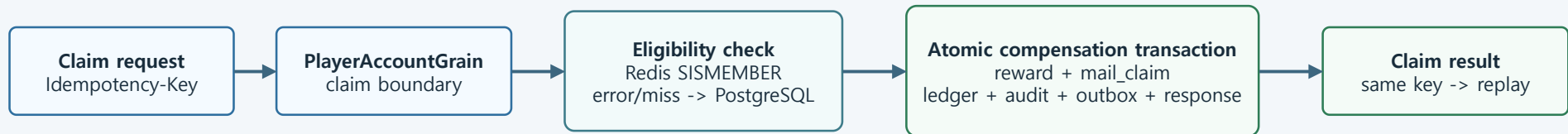
## 8.5 사고 복구와 보상

잘못된 콘텐츠를 되돌리는 것과 이미 영향을 받은 플레이어를 보상하는 것을 별도 단계로 처리한다.

### 운영 사고 대응



### 플레이어 보상 수령

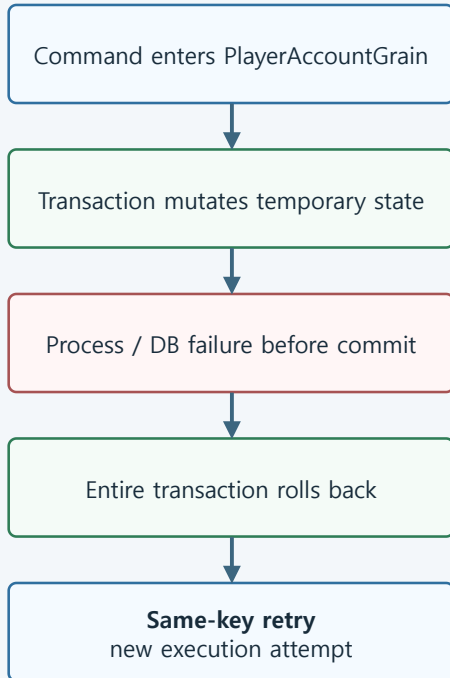


Redis is never the compensation source of truth; connection failure must fall back to PostgreSQL.

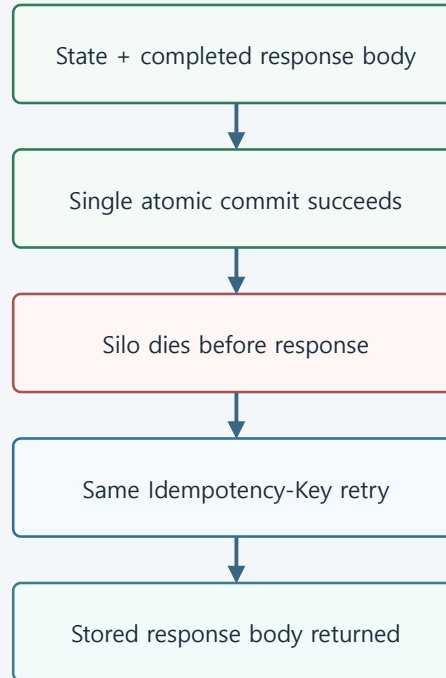
## 8.6 멍등성과 장애 복구

장애 시점을 분리해 rollback, response replay, Outbox retry의 책임을 명확히 한다.

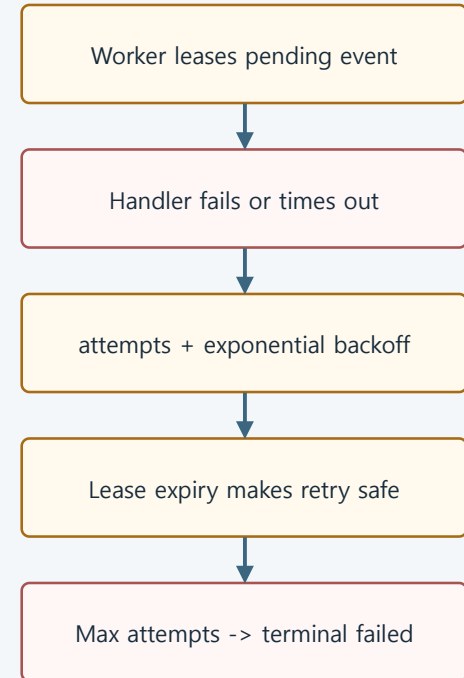
### A. Commit 전 장애



### B. Commit 후 응답 전 장애



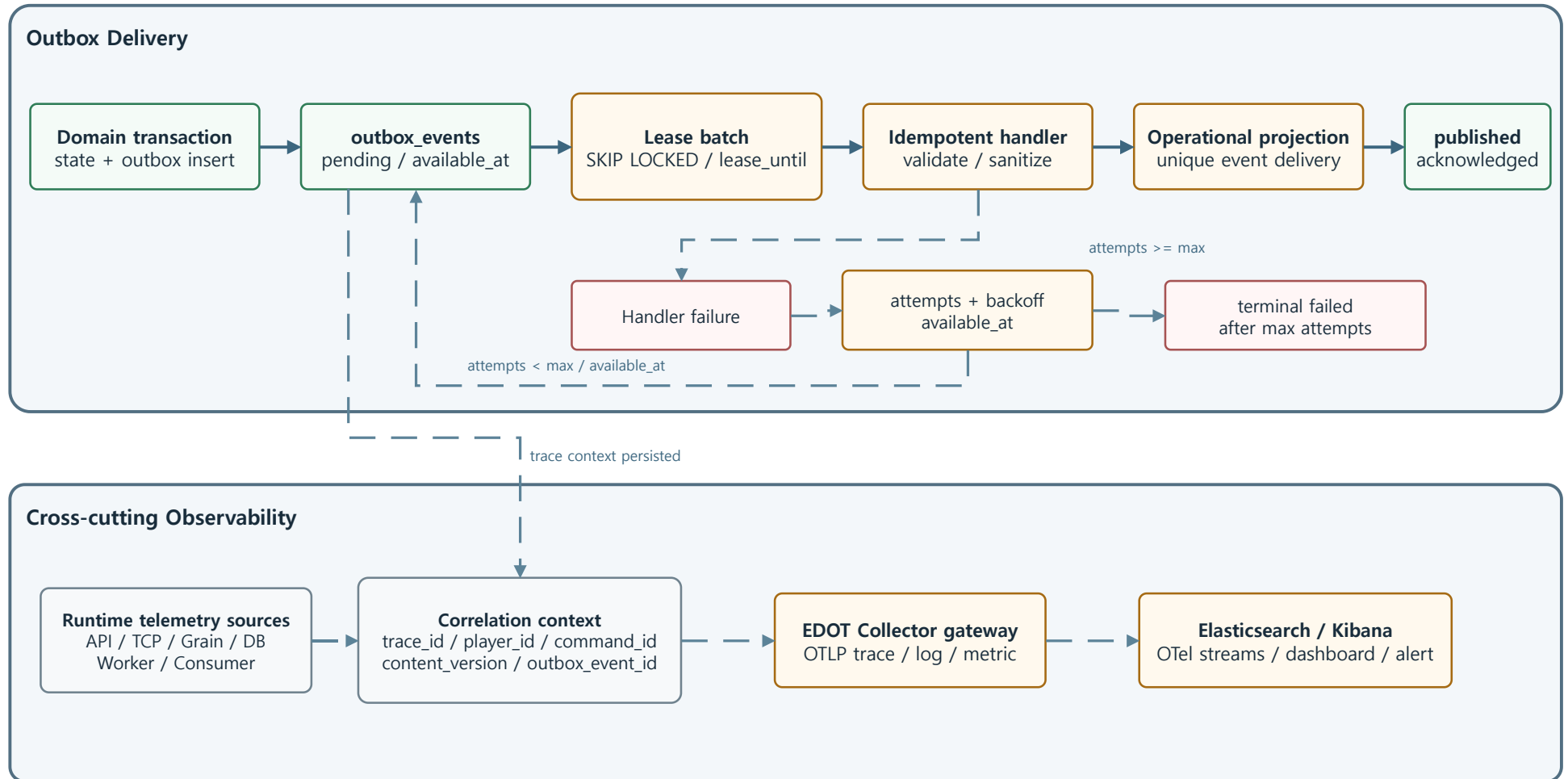
### C. Outbox 처리 장애



Request idempotency: COMPLETED only, no PENDING | Outbox processing state: pending / processing / published / failed

## 8.7 Outbox와 관측성

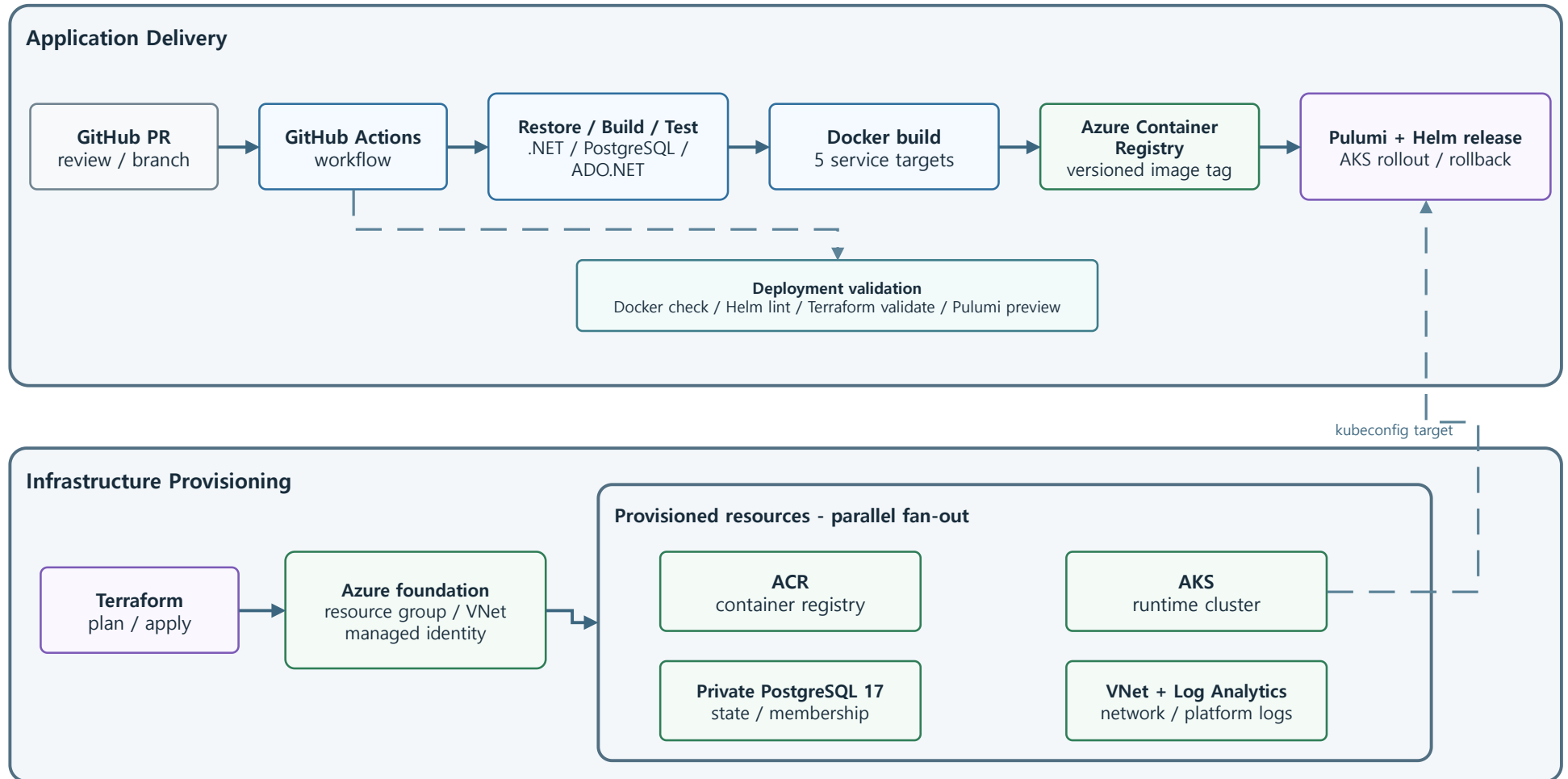
Outbox는 idempotent consumer로 전달하고, Observability는 모든 실행 단계를 횡단해 관찰한다.



Persist trace context with the Outbox event so Worker spans can be linked to the originating command.

## 8.8 CI/CD와 Azure 배포

런타임 요청 흐름과 배포/IaC 흐름을 분리해 배포 책임과 검증 경계를 명확하게 보여준다.



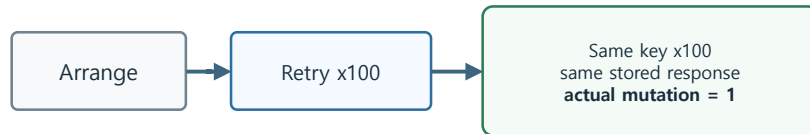
Local: Docker Compose | CI: GitHub Actions | Registry: ACR | Runtime: AKS | IaC: Terraform | App release: Pulumi + Helm



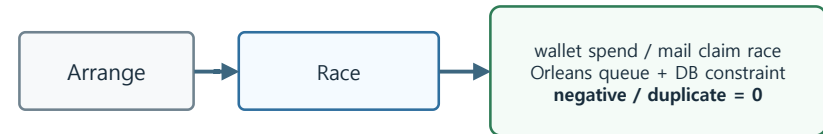
## 8.9 핵심 검증 시나리오

실패·동시성·복구 테스트로 아키텍처의 동작을 검증한다.

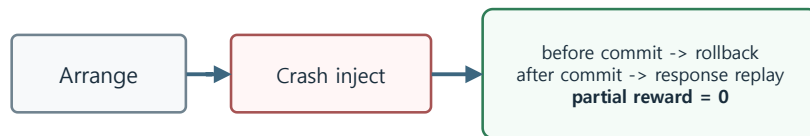
### 1. Idempotency Replay



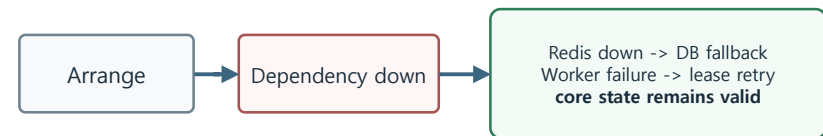
### 2. Concurrent Player Commands



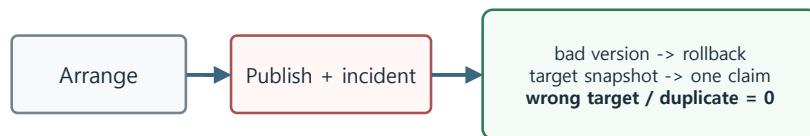
### 3. Crash Windows



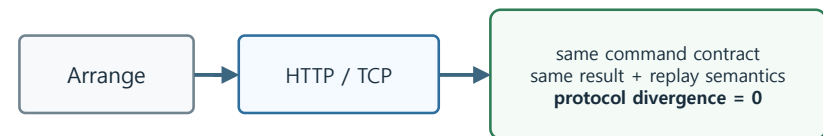
### 4. Dependency Failure



### 5. Content Rollback + Compensation



### 6. HTTP / TCP Command Parity



**Verification evidence** - xUnit 91 / PostgreSQL 17 / ADO.NET 2-Silo + E2E 4 / Docker 5 targets / Helm lint / Terraform validate / Pulumi preview

Tests must fail visibly when dependencies are not available; gated tests must not silently return as passed.